

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
**Department of Computer Science and Engineering**  
**ASSESSMENT TOOL 2 – APPLICATION BASED QUESTIONS (5 Questions)**

|                  |                                                                                                                          |               |                                                 |
|------------------|--------------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------------------|
| Course Code:     | DSA0607                                                                                                                  | Course Title: | Data Handling and Visualization                 |
| CO Assessed:     | CO2 – Precision (BL3)                                                                                                    | Assessment:   | Assessment Tool 2 – Application Based Questions |
| Weightage:       | 20% of CIA                                                                                                               | Total Marks:  | 25 (5 Questions × 5 Marks)                      |
| CO1 Statement:   | <i>Apply appropriate visualization methods to represent distributions, proportions, and relationships in data (BL3).</i> |               |                                                 |
| Student Name:    | B Akshaya                                                                                                                | Reg. No.:     | 192424336                                       |
| Section / Batch: | D/2024                                                                                                                   | Date:         | 25.07.2026                                      |

**Code of Conduct**

*I B Akshaya(192424336) Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Akshaya

**Instructions**

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

| Section / Topic                 | Focus                                                                        | Q Nos | Marks    |
|---------------------------------|------------------------------------------------------------------------------|-------|----------|
| Amounts & Proportions           | Category-wise comparison and proportion analysis using bar charts            | 1     | 5        |
| Distributions                   | Analyzing data spread and patterns using histograms, density plots, boxplots | 2     | 5        |
| Temporal & x–y Relationships    | Studying trends over time and relationships between variables                | 3     | 5        |
| Geospatial & Temporal Data      | Mapping locations and analyzing delivery patterns over time                  | 4     | 5        |
| Multivariate Data Visualization | Exploring multiple variables and comparing distributions                     | 5     | 5        |
| GRAND TOTAL:                    |                                                                              |       | 25 Marks |

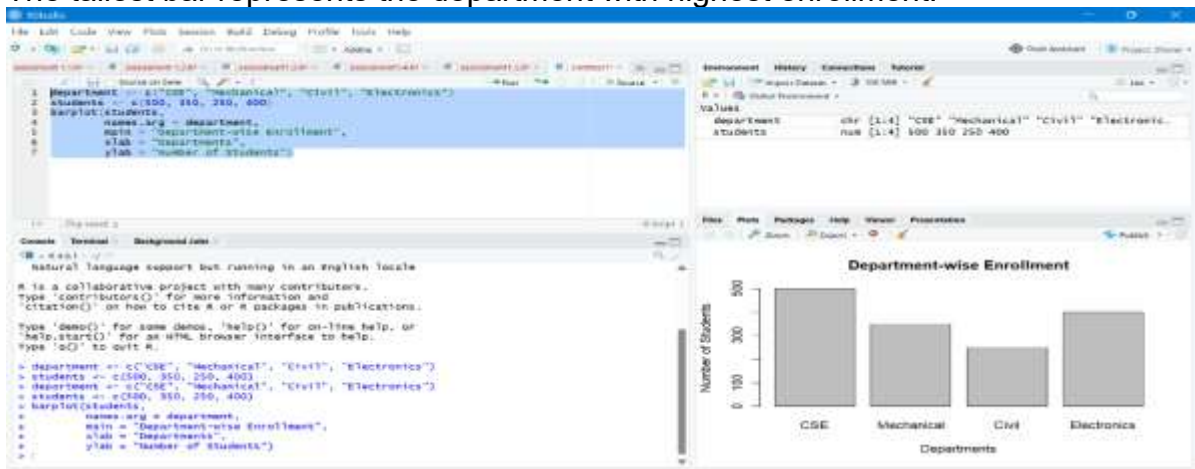
## Annexure A – Application Based Questions

1.A university wants to analyse student enrollments across various departments such as Computer Science, Mechanical Engineering, Civil Engineering, and Electronics. The administration aims to identify departments with the highest enrollments, compare enrollment trends over multiple semesters, and understand the proportion of students in each department. How would you categorize the data and choose appropriate visualizations to represent enrollment amounts, proportions, and temporal trends across departments?

Appropriate Visualizations

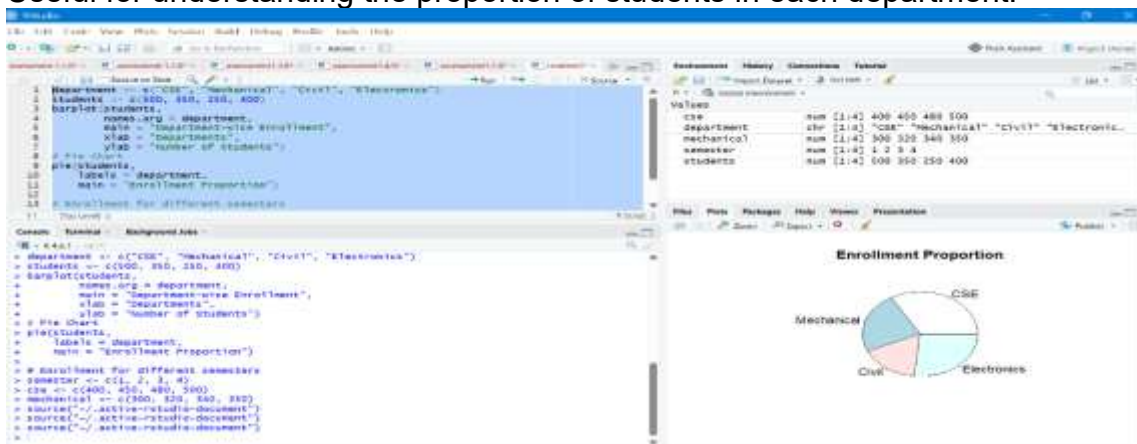
Bar Chart – Enrollment Amounts

- X-axis → Departments
- Y-axis → Number of students
- Used to compare enrollment among departments.
- The tallest bar represents the department with highest enrollment.



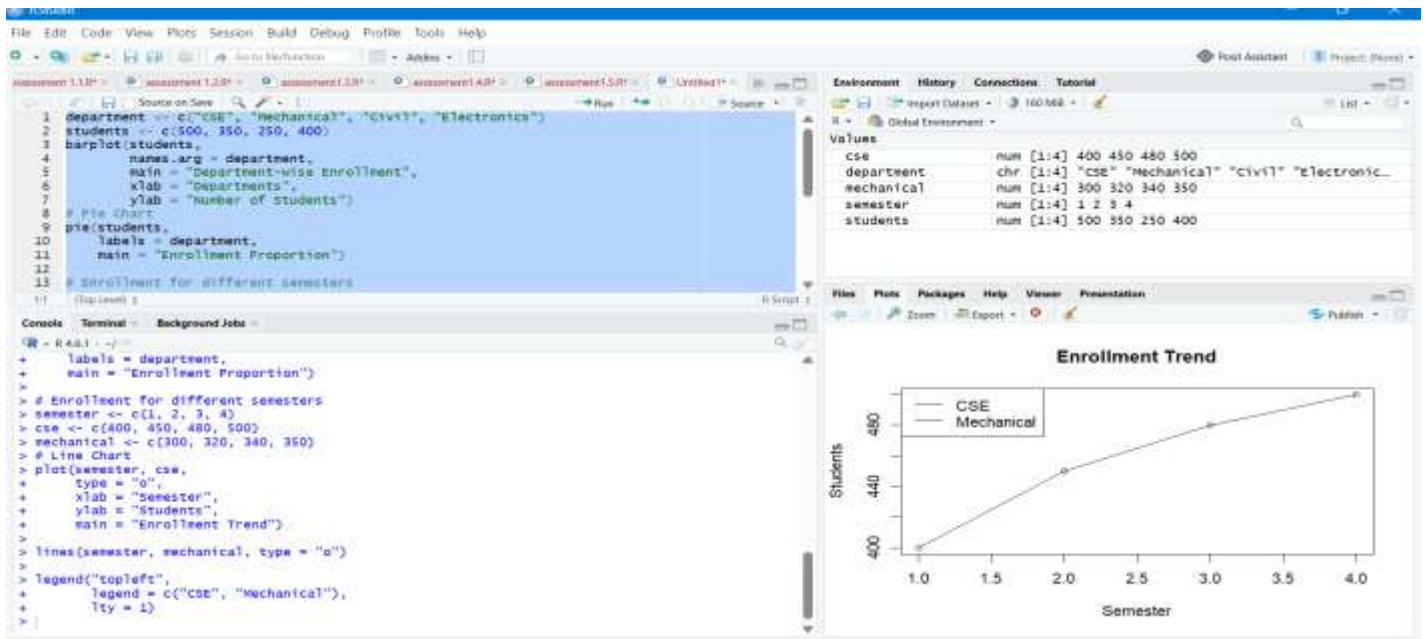
Pie Chart – Enrollment Proportions

- Each slice represents one department.
- Size of each slice shows its percentage of total enrollment.
- Useful for understanding the proportion of students in each department.



Line Chart – Enrollment Trends

- X-axis → Semesters
- Y-axis → Number of students
- Separate line for each department.
- Shows whether enrollment increases or decreases over time



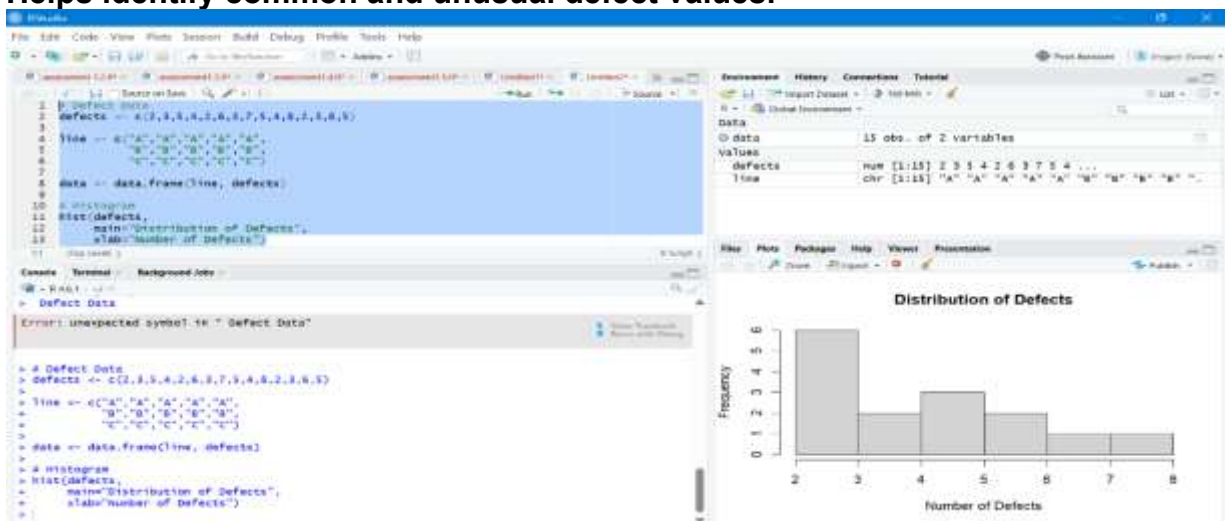
## Conclusion

Therefore, Bar Chart is suitable for comparing enrollment amounts, Pie Chart for showing proportions, and Line Chart for analyzing enrollment trends across multiple semesters.

**2.A manufacturing company records the number of defects found in products produced by different production lines. The quality assurance team wants to examine the distribution of defects, identify variability among production units, and detect unusual patterns. How would you use histograms, density plots, violin plots, and boxplots to analyze defect distributions and compare quality performance across production lines?**

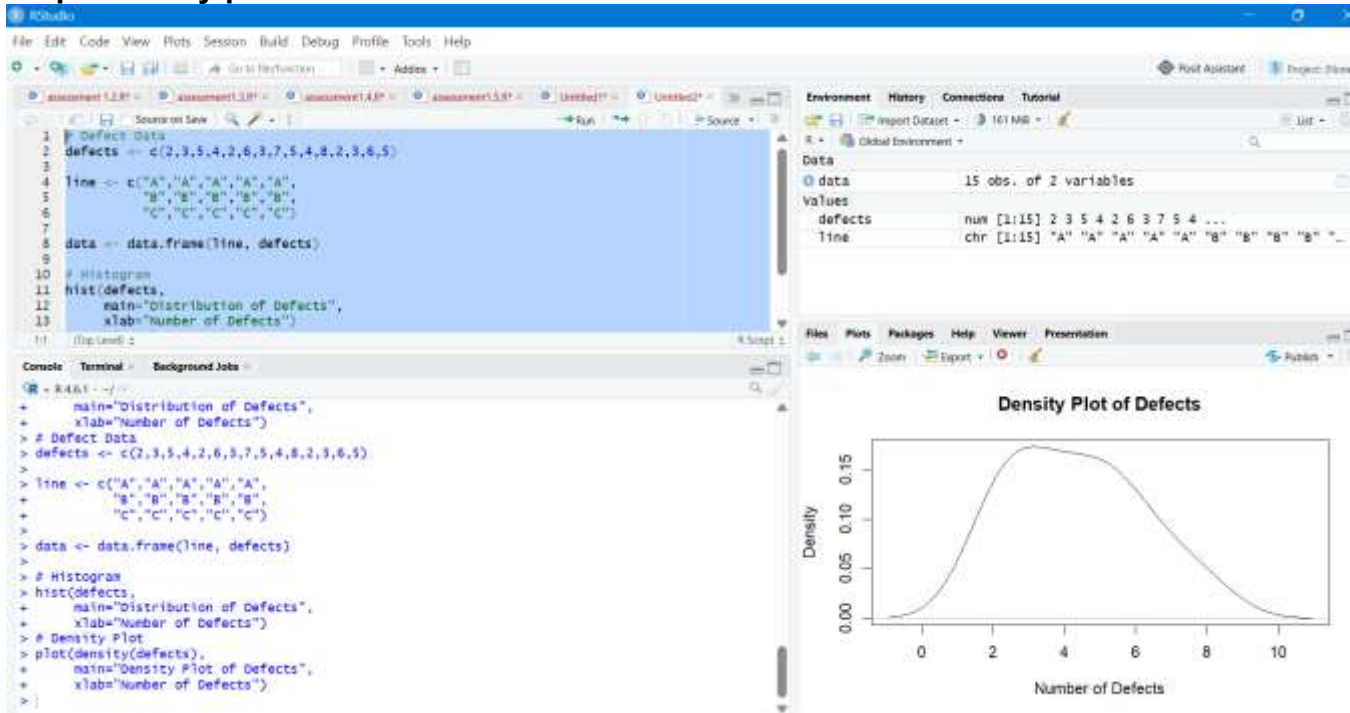
## Histogram – Defect Distribution

- X-axis → Number of defects
- Y-axis → Frequency
- Shows how frequently different defect counts occur.
- Helps identify common and unusual defect values.



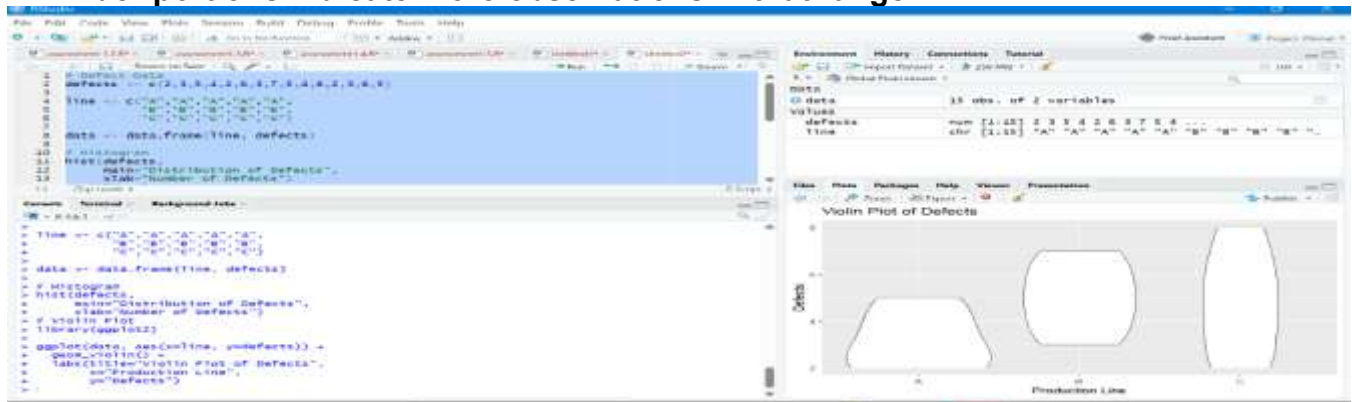
## Density Plot – Distribution Shape

- X-axis → Number of defects
- Y-axis → Density
- Shows the overall shape of defect distribution.
- Helps identify peaks and concentration of defects.



## Violin Plot – Compare Production Lines

- X-axis → Production Lines
- Y-axis → Number of defects
- Shows the distribution and spread of defects.
- Wider portions indicate more observations in that range.
- Wider portions indicate more observations in that range.



## Boxplot – Variability and Outliers

- Wider portions indicate more observations in that range.
- Boxplot – Variability and Outliers
- Boxplot – Variability and Outliers
- X-axis → Production Lines
- Y-axis → Number of defects
- Shows median, quartiles, and variability.
- Helps identify unusual values or outliers.

**3. An online learning platform tracks student activity including time spent on courses, number of videos watched, quiz scores, and course completion rates. The platform seeks to understand relationships among these variables and identify factors influencing successful course completion. How would you use scatter plots, heatmaps, and multivariate visualizations to analyse relationships among learner engagement metrics and academic outcomes?**

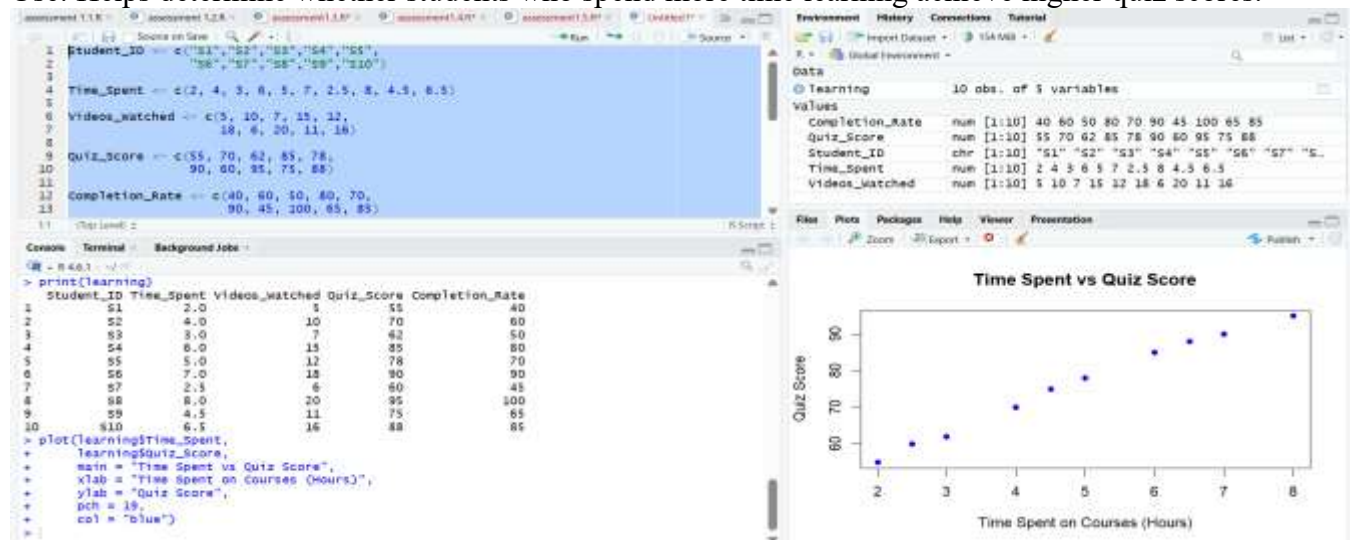
**Aim:** An online learning platform can use scatter plots, heatmaps, and multivariate visualizations to understand relationships between student engagement and academic outcomes such as quiz scores and course completion rates.

### 1. Scatter Plot

A scatter plot can analyse the relationship between study time and quiz scores.

- X-axis → Time Spent on Courses
- Y-axis → Quiz Score
- Each point → One student
- Trend line → Overall relationship

Use: Helps determine whether students who spend more time learning achieve higher quiz scores.

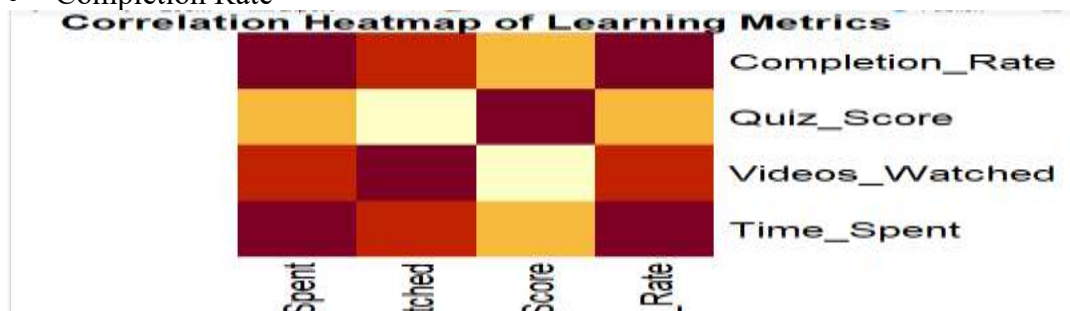


### 2. Heatmap

A correlation heatmap can compare all important numerical variables.

Variables can include:

- Time Spent
- Videos Watched
- Quiz Score
- Completion Rate

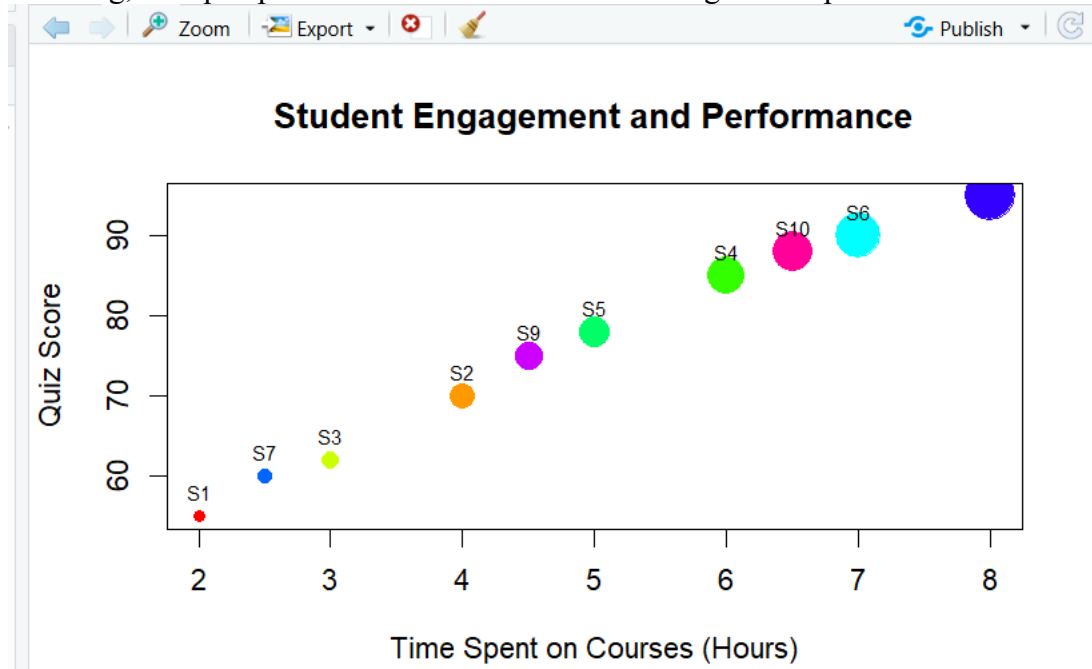


### 3. Multivariate Visualization

A bubble plot can display several variables simultaneously.

- X-axis → Time Spent
- Y-axis → Quiz Score
- Bubble size → Videos Watched
- Color → Course Completion Rate

Use: Helps identify highly engaged students and determine whether combinations of study time, video watching, and quiz performance are associated with higher completion rates.



#### Conclusion

These visualizations help identify student engagement patterns, factors affecting quiz performance, and variables associated with successful course completion. The platform can use these findings to improve course content, recommend learning resources, identify students requiring additional support, and develop effective student-engagement strategies.

**4.A city administration collects traffic data including vehicle counts, congestion levels, accident locations, and travel times across different regions and time periods. The objective is to identify traffic hotspots and improve transportation planning. How would you represent geospatial and temporal data using maps, heatmaps, and time-series visualizations to identify traffic patterns and optimize road management?**

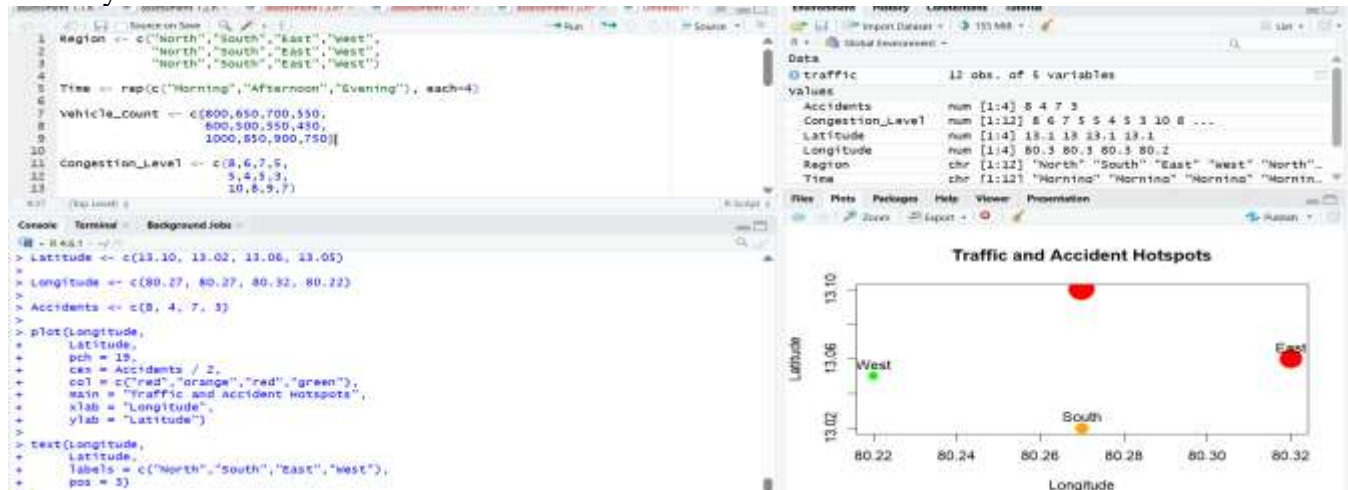
**Aim:** A city administration can use geospatial maps, heatmaps, and time-series visualizations to analyze vehicle counts, congestion levels, accident locations, and travel times across different regions and time periods.

#### 1. Geospatial Map

A geospatial map can display traffic and accident information according to geographic location.

- Latitude and Longitude → Location
- Point size → Vehicle Count
- Colour → Congestion Level
- Points → Accident Locations

Use: Helps identify traffic and accident hotspots and shows which regions require better road management or safety measures.

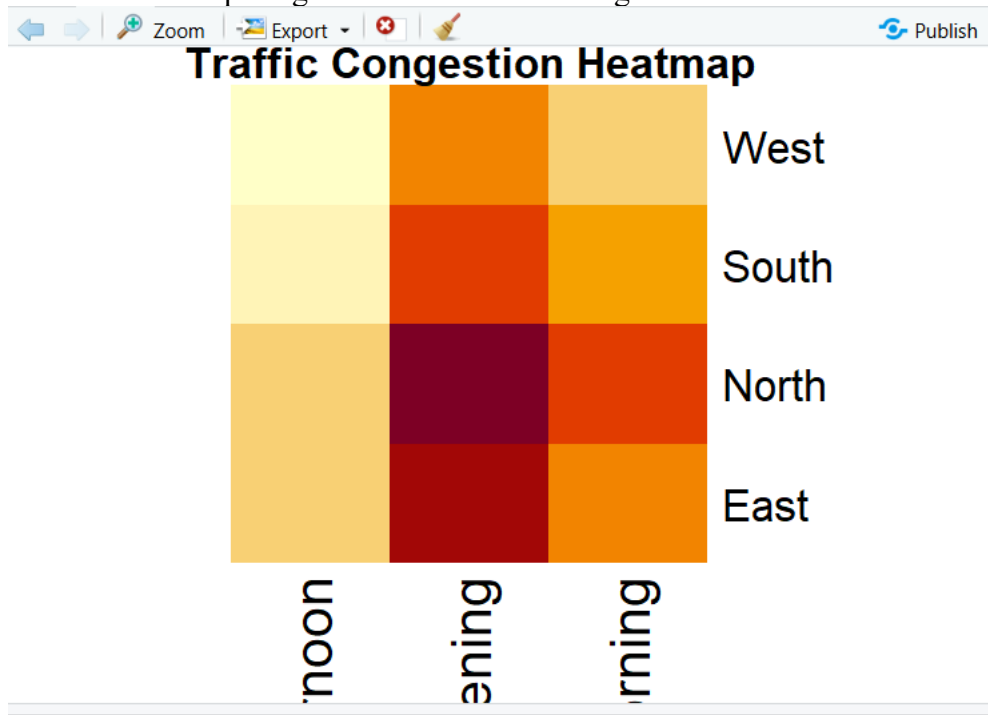


## 2. Heatmap

A heatmap can represent congestion levels across different regions and time periods.

- Rows → Regions
- Columns → Time Period
- Color intensity → Congestion Level

Use: Helps identify highly congested regions and peak traffic periods. Darker or stronger colors can indicate areas requiring immediate traffic management.

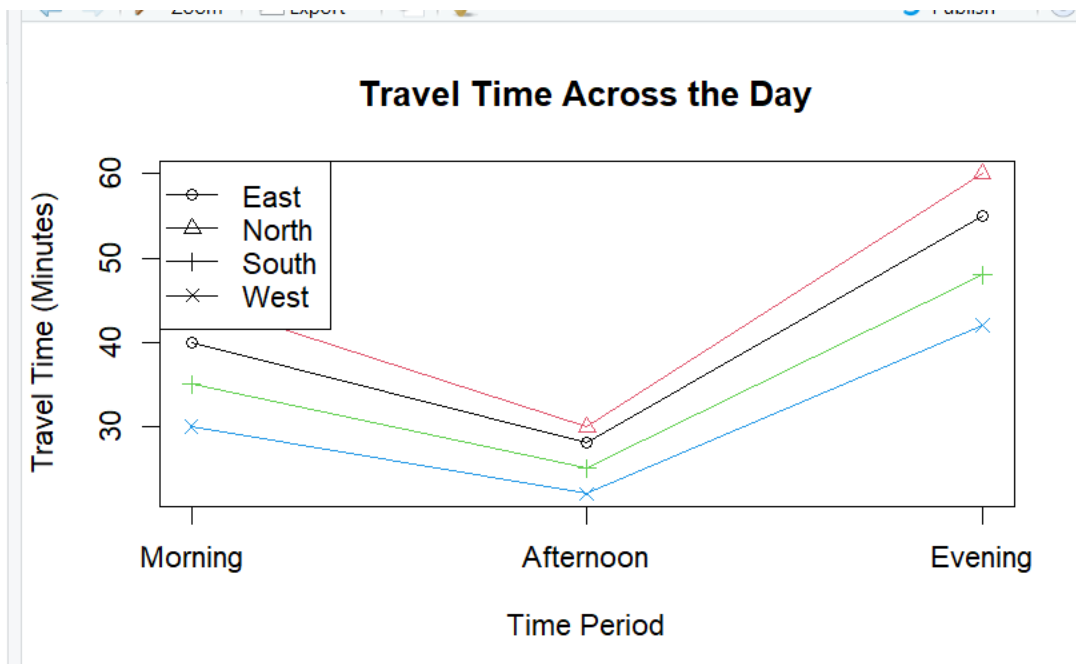


## 3. Time-Series Visualization

A line chart can show how vehicle counts or travel times change throughout the day.

- X-axis → Time Period
- Y-axis → Vehicle Count
- Lines → Different Regions

Use: Helps identify morning and evening peak hours and understand how traffic conditions change over time.



#### Conclusion

These visualizations help city administrators identify congestion hotspots, accident-prone areas, peak traffic periods, and inefficient routes. The results can support traffic-signal optimization, alternative-route planning, road expansion, accident prevention, and long-term transportation infrastructure planning.

**5.A large organization records communication data among employees through emails, meetings, and project collaborations. Management wants to identify key employees, collaboration groups, and communication bottlenecks within the organization. How would you use network visualizations and multivariate analysis techniques to represent communication patterns, identify influential individuals, and improve organizational collaboration?**

**Aim:** A large organization can use network visualizations and multivariate analysis techniques to understand communication through emails, meetings, and project collaborations. These methods help identify influential employees, collaboration groups, and communication bottlenecks.

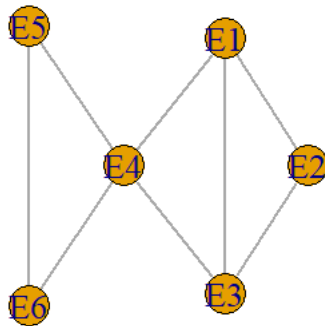
#### 1. Network Visualization

A network graph can represent communication relationships among employees.

- Nodes → Employees
- Edges → Communication between employees
- Node size → Number of connections
- Edge width → Communication frequency

**Use:** Helps identify highly connected and influential employees. Employees with many connections may act as important communication hubs within the organization.

## Employee Communication Network

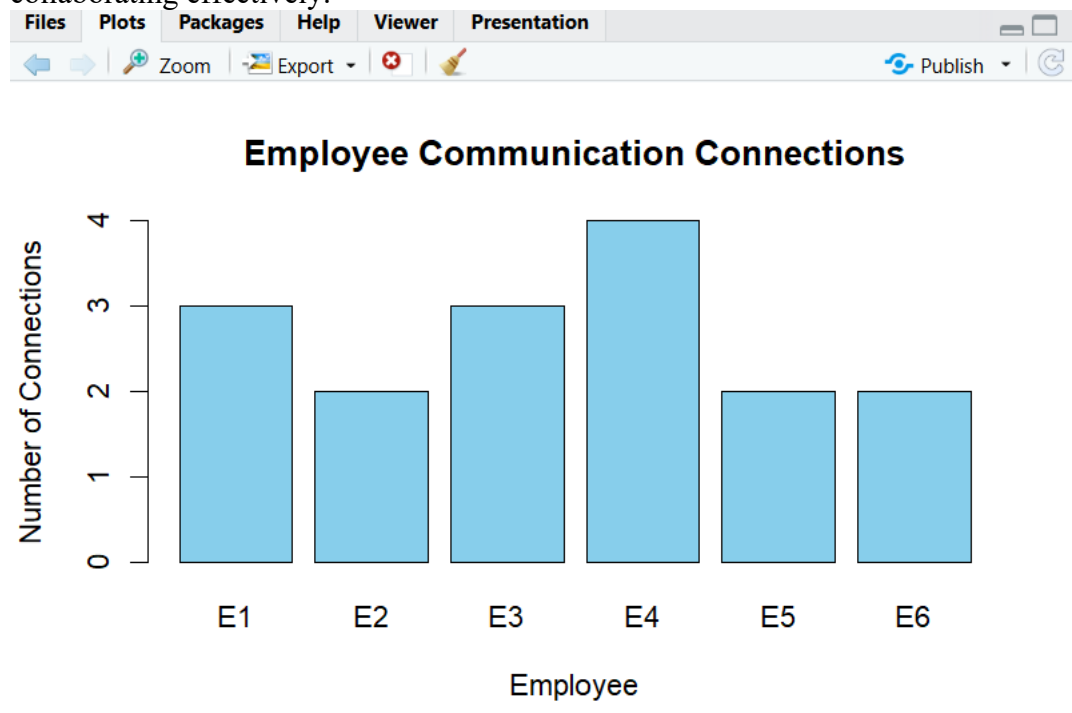


### 2. Collaboration Group Analysis

Network visualization can also reveal groups of employees who communicate frequently with one another.

- Closely connected nodes → Collaboration groups
- Different groups → Departments or project teams

Use: Helps management identify strong teams and determine whether different departments are collaborating effectively.



### 3. Communication Bottleneck Analysis

Employees who connect otherwise separate groups may act as communication bridges.

Use: If too much information flows through only one or two employees, they may become communication bottlenecks. Management can create additional communication channels to improve information flow.

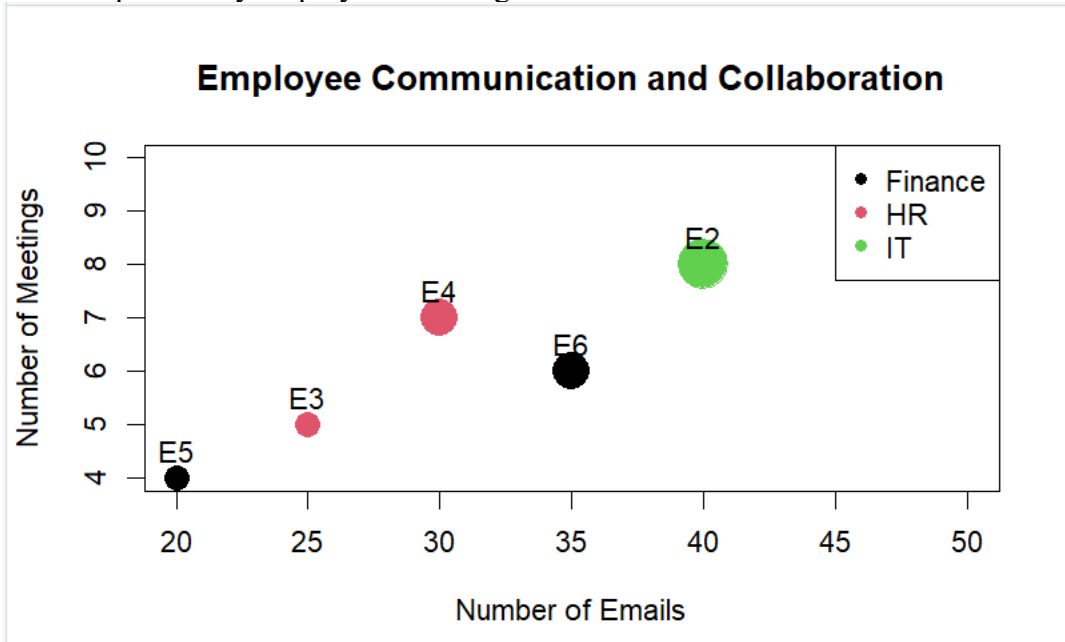
### 4. Multivariate Visualization

A bubble plot can compare several employee communication variables simultaneously.

- X-axis → Emails
- Y-axis → Meetings
- Bubble size → Project Collaborations

- Color → Department

Use: Helps identify employees with high overall communication and collaboration activity.



#### Conclusion

These visualizations help management identify influential employees, collaboration groups, isolated employees, and communication bottlenecks. The results can support better team formation, cross-department communication, workload distribution, and organizational collaboration.

## Rubrics for Calculation

| Criteria                         | Weight age | Level 4 – Exemplary                                                        | Level 3 – Proficient                                  | Level 2 – Developing                           | Level 1 – Beginning                                    | Marks |  |
|----------------------------------|------------|----------------------------------------------------------------------------|-------------------------------------------------------|------------------------------------------------|--------------------------------------------------------|-------|--|
| Understanding of Problem Context | 20%        | Clearly understands the problem and accurately identifies all requirements | Good understanding with minor gaps                    | Basic understanding; some requirements unclear | Poor understanding or misinterpretation of the problem |       |  |
| Application of Concepts          | 30%        | Accurately applies appropriate concepts to solve complex problems          | Applies relevant concepts correctly with minor errors | Applies basic concepts with limited accuracy   | Fails to apply appropriate concepts                    |       |  |
| Problem-Solving Approach         | 20%        | Logical, well-structured, and efficient approach                           | Mostly logical approach with minor gaps               | Basic approach with some inconsistencies       | Illogical or unclear approach                          |       |  |
| Accuracy of Solution             | 20%        | Completely correct solution with proper justification                      | Mostly correct with minor errors                      | Partially correct solution                     | Incorrect or incomplete solution                       |       |  |
| Clarity                          | 10%        | Clear, well-organized, and properly explained solution                     | Understandable with minor clarity issues              | Limited clarity and explanation                | Poorly presented and difficult to understand           |       |  |

| Faculty In-charge | Course Coordinator | Head of Department |
|-------------------|--------------------|--------------------|
|                   |                    |                    |
| Signature: _____  | Signature: _____   | Signature: _____   |
| Date: _____       | Date: _____        | Date: _____        |